

18.06 Linear Algebra - Lectures 25, 26, 27

Textbook Notes - Sections 6.5, 9.2, 9.3

• Section 6.5 Positive Definite Matrices

Symmetric matrices with all positive eigenvalues.
Positive eigenvalues ties together pivots, determinants, eigenvalue, energy, and least squares.

• The Five Equivalent Tests

For a symmetric matrix S , these are all true or all false together:

1. All n pivots are positive
2. All n upper-left determinants are positive
3. All n eigenvalues are positive
4. $x^T S x > 0$ for every $x \neq 0$ (energy test)
5. $S = A^T A$ for some A with independent columns

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

↳ Any one test is decisive

↳ Test 2 (determinants) usually fastest by hand

↳ Test 1 (pivots) fastest computationally.

• 2x2 Shortcut

For $S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$

$$\begin{aligned} &\rightarrow a > 0 \\ &\rightarrow ac - b^2 > 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} &\rightarrow a > 0 \\ &\rightarrow ac - b^2 > 0 \end{aligned}} \right\} \&$$

↳ Pivots are (1) a (2) $(ac - b^2)/a$

• Energy Definition

$$x^T S x = ax^2 + 2bxy + cy^2$$

- The diagonal entries produce the pure squares
- The off-diagonal entries produce the cross terms (counted twice)

⇒ Consequence : if S and T are positive definite, so is $S+T$, since:
$$x^T(S+T)x = x^T S x + x^T T x$$

↳ Energies add cleanly (pivots & eigenvalues do not).

• Why $S = A^T A$ works?

$$x^T(A^T A)x = (Ax)^T(Ax) = \|Ax\|^2$$

↳ Positive whenever $Ax \neq 0$, which is guaranteed when A has independent columns.

↳ Links Positive Definiteness directly to Least Squares from Chapter 4.

• Three Factorizations (three ways to write the energy as squares)

$$S = A^T A \rightarrow \|Ax\|^2 \quad \text{from model/difference matrix}$$

$$S = LDL^T \rightarrow \sum d_i \cdot (\text{pivot combinations})^2 \quad \text{Cholesky } A = LU$$

$$S = Q \Lambda Q^T \rightarrow \sum \lambda_i (q_i^T x)^2 \quad \text{Eigenvalues + vectors}$$

→ Each expresses $x^T S x$ as a sum of squares with positive coefficients (exactly what positive definiteness is).

• Positive Semidefinite

The boundary case: $\lambda \geq 0$, pivots ≥ 0 ,
determinants ≥ 0 , $x^T S x \geq 0$

- ↳ Matrix is singular $\rightarrow$ some eigenvalue is exactly zero
- $\rightarrow$ eigenvector has zero energy
- $\rightarrow$ Still $S = A^T A$, but A has dependent columns.

• The Ellipse $x^T S x = 1$:

The graph is an ellipse (ellipsoid in R^n) when S is positive definite

- $\rightarrow$ axes point along the eigenvectors of S
- $\rightarrow$ half-lengths are $\frac{1}{\sqrt{\lambda_i}}$

• The larger eigenvalue gives the shorter axis.

↳ $S = Q \Lambda Q^T$ is called the Principle Axis Theorem

↳ Q rotates the tilted ellipse into alignment with the coordinate axes, where the equation becomes:

$$\lambda_1 X^2 + \lambda_2 Y^2 = 1$$

- If one eigenvalue is negative $\rightarrow$ graph becomes a Hyperbola
- If both are negative $\rightarrow$ no real points at all.

• Test for a Minimum

For $F(x, y)$, the second-derivative matrix is:

$$S = \begin{bmatrix} \partial^2 F / \partial x^2 & \partial^2 F / \partial x \partial y \\ \partial^2 F / \partial y \partial x & \partial^2 F / \partial y^2 \end{bmatrix}$$

- F has a minimum at a point where the first derivatives vanish, and S is positive definite there. It generalises the single-variable test: $f' = 0$ and $f'' > 0$.

◦ Section 9.2 Hermitian and Unitary Matrices

Every real concept carries over to complex matrices by replacing transpose with conjugate transpose.

• Conjugate Transpose

$$A^H = \bar{A}^T \quad (\text{transpose and conjugate every entry})$$

Rules that carry over directly:

$$\hookrightarrow (AB)^H = B^H A^H$$

$$\hookrightarrow \langle Au, v \rangle = \langle u, A^H v \rangle \quad (A^H \text{ is the adjoint of } A)$$

• Why the conjugate is necessary?

For a complex vector, $z^T z$ fails as a length:

$$z = (1, i) \rightarrow z^T z = 1 + i^2 = 0$$

(nonzero vector, zero length)

The fix uses $\bar{z}z = |z|^2$:

$$\|z\|^2 = z^H z = |z_1|^2 + \dots + |z_n|^2$$

(always real and positive)

Inner product: ~~$\langle u, v \rangle$~~ $= u^H v$

Order matters $v^H u$ is the conjugate of $u^H v$

• Hermitian Matrices $S = S^H$

The complex analogue of symmetric.

Entry condition: $S_{ij} = \bar{S}_{ji}$ (forcing the diagonal to be real)

• Three Properties

1. $z^H S z$ is real for every z (the "energy" is real)
2. All eigenvalues are real ($\lambda = z^H S z / z^H z$, ratio of reals)
3. Eigenvectors for distinct λ are orthogonal

→ Every real symmetric matrix is Hermitian (conjugating does nothing)

• Unitary Matrices $Q^H Q = I$

Complex analogue of orthogonal.

Columns are orthonormal under the complex inner product;
for square Q , $Q^H = Q^{-1}$

$$\hookrightarrow \|Qz\| = \|z\| \quad (\text{lengths preserved})$$

$$\hookrightarrow |\lambda| = 1 \quad (\text{all eigenvalues on the unit circle})$$

• Correspondence

Real	Complex	Condition
Symmetric	Hermitian	$S = S^H$, real λ
Orthogonal	Unitary	$Q^H Q = I$, $ \lambda = 1$
$S = Q \Lambda Q^T$	$S = Q \Lambda Q^H$	Spectral Theorem

→ Normal matrices are the general ^{class} ~~case~~ with orthonormal eigenvectors
 $\hookrightarrow N N^H = N^H N$

→ Hermitian, skew-Hermitian, and Unitary matrices are all normal.

→ Four Fundamental subspaces carry over too → $C(A^T)$ becomes $C(A^H)$
 and orthogonality is with respect to the complex inner product.

Section 9.3 Fast Fourier Transform

The Fourier matrix F_n has no zero ~~matrix~~^{entries}, so multiplying by it appears to cost n^2 operations.

→ Its special structure permits a recursive factorization
↳ Reducing cost to $\frac{1}{2}n \log_2 n$

• Roots of Unity

The solutions to $z^n = 1$ are n evenly spaced points on the unit circle:

$$w = e^{(2\pi i/n)} \quad \text{with } 1, w, w^2, \dots, w^{n-1}$$

Key properties: $w^n = 1$, and powers sum to zero
 $1 + w + w^2 + \dots + w^{n-1} = 0$
by symmetry.

• The Fourier Matrix

The entry in row j , column k is $w^{(jk)}$, with ~~indices~~^{indices} running from 0 to $n-1$:

$$F_n = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & w & w^2 & \dots & w^{n-1} \\ 1 & w^2 & w^4 & \dots & w^{2(n-1)} \\ \dots & \dots & \dots & \dots & \dots \\ 1 & w^{n-1} & w^{2(n-1)} & \dots & w^{(n-1)^2} \end{bmatrix}$$

→ Properties

- F_n is symmetric but not Hermitian (conjugating $+i$ to $-i$)
- Columns are orthogonal with $F^* F = nI$, so $F^{-1} = \frac{F^*}{n} = \frac{1}{n} \bar{F}^T$
- $\frac{F}{\sqrt{n}}$ is unitary
- The inverse is the same matrix with w replaced by $\bar{w}$ (sign i changes)

• Interpretation : $F_c = y$ evaluates the Fourier series

$$\sum C_k e^{(ikw)} \text{ at } n \text{ equally spaced points.}$$

- F moves from frequency space to physical space
- F^{-1} recovers the coefficients.

• FFT Factorization

- The key is that $W_n^2 = W_{n/2}$
 - ↳ squaring the n th root gives the $(n/2)$ th root
 - ↳ connects F_n to 2x copies of $F_{n/2}$

$$F_n = \begin{bmatrix} I & 0 \\ I & -D \end{bmatrix} \begin{bmatrix} F_{n/2} & 0 \\ 0 & F_{n/2} \end{bmatrix} \begin{bmatrix} \text{even-odd} \\ \text{permutation} \end{bmatrix}$$

↳ where $D = \text{diag}(1, w, w^2, \dots, w^{n-1})$ with $m = n/2$.

• Three Stages :

1. Permute $\rightarrow$ separate C into even-indexed C' and odd-indexed C''
2. Transform $\rightarrow$ apply the half-size $F_{n/2}$ to each part.
3. Combine $\rightarrow y_j = y_j' + w^j y_j''$ and $y_{j+n} = y_j' - w^j y_j''$

• Why the cost drops :

- Each step halves the problem size at the cost of $\frac{n}{2}$ multiplications for the diagonal D .
- Recursing $\log_2 n$ times :

$$n^2 \rightarrow \frac{1}{2} n \log_2 n$$

• For $n = 1024$: roughly 1 million operations become ~ 5000 ^{feasible}
 ↳ Reason why digital signal processing, image compression, etc.